is a city where 40 million litres of sewage waste collected every day is treated at tertiary level and brought back in the city for non-potable usage—for example, use of industrial waste water for irrigation. For areas of the city that are not connected to the main sewage line, we have set up local plants that collect sewage from the areas, treat it and give it back to the local inhabitants for reuse. This is our Zero Liquid Discharge project, and we believe that sustainable technologies like this should be taken up by every city to see long-term improvement."

Mathur put forward more real-time waste management solutions as examples, "We have patented a technology where all the roads we construct are built using waste plastics. Moreover, Jamshedpur has more than 15 biogas stations that recycle biodegradable waste to generate usable biogas."

Jamshedpur has also adopted sensorised bins, which track the level of waste collection in real time. JUSCO is on the verge of activating LoRaWAN connectivity for better maintenance of these smart systems.

Many solution providers are suggesting waste collection below the ground. An example is set by the Gujarat International Finance-Tec (GIFT) City—a business district in Gujarat which has been included as a model city in smart city campaign. It has introduced the country's first automated waste collection infrastructure. Instead of trash bins, it uses inlets for waste collection. The inlets connect to large underground suction pipe networks that end up into a central collection hub. Any amount of garbage put through the inlets is sucked through the pipes into the central hub. At the collection inlet, the waste is separated into biodegradables and non-biodegradables.

The project is under expansion and has resulted in no open waste and garbage in the city, no unpicked waste piles in an area and no manual intervention for waste collection. The infrastructure won multiple awards at *CMO Asia 2017* in Singapore.

Countries throughout the world are using unique techniques to manage their wastes. Australia has installed two types of smart bins: Big Belly and Smart Belly. These bins can detect their waste-collection level in real time, segregate reusable waste from non-reusable waste at the point of collection, and compost biodegradable waste on the spot.

Colombia has come up with an exciting idea to engage its citizens in waste-recycling process. To promote return of reusable wastes like plastic bottles and bags, it has introduced Ecobot reverse vending machines. Each time citizens return reusable waste to this bin, the machine awards them with vouchers or coupons that can be used for shopping, movies and so on. This project is running successfully, attracting more citizens towards going green! EFY

—Paromik Chakraborty, technical journalist, EFY